

DSO 576 Module 6 - Project Data Cleaning

Project: Spotify Financial Performance Data Cleaning

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GitHub branch: branch_haley

Branch URL: (https://github.com/04Shreya13/Stock_Movement_Context_Dashboard/tree/branch_haley)

14-item checklist

1. State the project vision and question - Done

Evidence: The vision is a financial analyst agent that uses Spotify's public financial statements to describe performance. The business question is: how have revenue growth, gross margin, operating profitability, net profitability, operating cash flow, and liquidity changed over time? The main comparisons are year-over-year revenue growth, gross margin, and operating margin. See README.md and data/processed/spotify_year_summary.csv.

2. Define one row - Done

Evidence: One cleaned row is one financial metric for one calendar year in EUR. record_id = calendar_year + metric + unit is unique. The script found 61 rows and 61 unique record IDs. See reports/cleaning_audit.json.

3. Preserve the source - Done

Evidence: Source: SEC Company Facts for Spotify Technology S.A. (CIK 1639920), downloaded 2026-10-03 from <https://data.sec.gov/api/xbrl/companyfacts/CIK0001639920.json>. The dated raw file is unchanged at data/raw/spotify_companyfacts_2026-10-03.json; SHA-256 is cc022970e9b216233a25843b42f2a8a528dd92feb38500f9759c0109fccf84e0. The source is loaded as JSON by scripts/clean_spotify_financials.py.

4. Inspect columns and types - Done

Evidence: Raw values are nested JSON. The script parses period_start, period_end, and filed_date as ISO dates for validation; confirms values are numeric; converts the calendar year to integer; and writes numeric value_eur. There were 0 failed date or numeric conversions. See tests/test_cleaning.py::test_types_dates_and_units and reports/cleaning_audit.json.

5. Standardize text - Done

Evidence: XBRL concepts such as ProfitLossFromOperatingActivities become stable snake_case labels such as operating_profit_loss. Unit and form text are stripped and uppercased (EUR, 20-F), and accession numbers remain strings so punctuation/leading characters are preserved. See METRICS and flatten_selected_facts() in the cleaning script.

6. Investigate duplicates - Done

Evidence: Among 152 selected source observations, there were 0 exact duplicate rows. There were 91 extra observations with repeated metric/year/unit business keys because annual filings repeat comparative years. For each repeated key, the script retained the latest filing and kept all decisions in reports/deduplication_audit.csv. This produced 61 unique rows. No repeated group had conflicting values.

7. Handle missing data - Done

Evidence: Before cleaning, 32 of 152 selected facts lacked `period_start` and 91 lacked `frame`; required end date, value, accession, and filed date had 0 missing values. After deduplication, 11 `period_start` values remain blank, all for `cash_and_cash_equivalents`. This is intentional because balance-sheet cash is an instant fact measured at year-end. `frame` is not carried into the analytic output because it is nonessential and systematically absent in comparative contexts. Every analytic value is present.

8. Check suspicious values - Done

Evidence: All dates are valid ISO dates; all filing dates occur after period end; duration facts span 365 or 366 days; all revenue is positive; and gross profit is between zero and revenue. Negative profit values were retained because they represent real losses. The unusually low 2022 operating cash flow (EUR 46 million) was checked and retained. See `test_financial_relationships_and_suspicious_values` and the record checks.

9. Verify merges - Not applicable

Explanation: No tables were merged. The input is one Company Facts document. The year summary pivots the cleaned long table by its already-validated year/metric key; each 2016-2025 year has all six metrics and `metric_value_count = 6`, so the pivot adds no records.

10. Reconcile changes - Done

Evidence: 152 selected observations to 61 cleaned rows. The difference is exactly 91 older repeated filing contexts removed by the latest-filing rule. No values were imputed and no unique annual facts were dropped. Important totals are not summed across filing contexts because doing so would triple-count comparative facts. `reports/deduplication_audit.csv` accounts for every selected row.

11. Check calculations - Done

Evidence: For 2016-2025, every annual metric has 10 nonmissing values (60 values total); the separate 2015 record is an opening cash balance. Gross margin denominator is revenue and uses 10 records; operating margin denominator is revenue and uses 10 records; revenue YoY uses 9 year pairs because 2016 has no prior-year revenue in the analysis table. Missing values are never treated as zero. Missing group keys are rejected by the required concept and date checks. See `create_year_summary()` and `spotify_year_summary.csv`.

12. Verify individual records - Done

Evidence: Five expectations were written in `create_record_checks()` and then compared with output. All 5 matched. The table below shows original/expected/actual values; full provenance is in `reports/five_record_checks.csv`.

Record	Original	Expected	Actual	Result
2017 net profit/loss	-1,235,000,000	-1,235,000,000	-1,235,000,000	Match
2021 operating profit/loss	94,000,000	94,000,000	94,000,000	Match
2022 operating cash flow	46,000,000	46,000,000	46,000,000	Match
2024 cash and equivalents	4,781,000,000	4,781,000,000	4,781,000,000	Match
2025 revenue	17,186,000,000	17,186,000,000	17,186,000,000	Match

13. Test reproducibility - Done

Evidence: From the project directory, run `python -m pip install -r requirements.txt`, `python scripts/clean_spotify_financials.py`, and `python -m unittest discover -s tests -v`. A clean run recreated all outputs and all 5 tests passed. `haley.ipynb` provides the same restart-and-run workflow.

14. Document remaining limitations - Done

Evidence: Company-reported XBRL is not an independent accounting audit. The selected six metrics exclude segment, subscriber, market-price, and acquisition-target data. Historical results describe financial capacity but

cannot alone recommend an M&A; decision. Latest-filing selection could incorporate a later restatement; therefore accession and filing date are retained, and all contexts remain in the audit. The next phase needs defined M&A; targets, valuation assumptions, and risk criteria.

Cleaning decision log

Rule	Reason	Check
Select six decision-relevant IFRS concepts	Keeps the dataset aligned with growth, profitability, cash generation, and liquidity	All six concepts exist; 61 final facts
Keep FY Form 20-F facts in EUR ending December 31	Makes years and units comparable	All final rows are 20-F/FY/EUR; 0 invalid dates
Require 365/366 days for duration facts	Prevents quarterly or year-to-date values from being mislabeled annual	Automated duration validation passes
Allow blank start only for instant cash facts	A balance-sheet fact is measured on a date, not over a duration	All 11 remaining blanks are cash instant facts
Keep latest filing for each metric/year/unit	Later filings are the best available presentation and repeated comparisons must not be double-counted	91 older contexts removed; 0 value conflicts
Preserve negative profit values	A loss is meaningful data, not an error	Negative cases remain and pass record checks
Do not impute financial values	Invented values would distort margins and trends	0 missing analytic values; no fill operation exists
Use explicit formulas and denominators	Makes derived metrics auditable	10 gross margins, 10 operating margins, 9 YoY growth rates

How I checked a Codex suggestion

Codex suggested deduplicating repeated annual facts by retaining the latest filing date. I did not accept that only from the suggestion. I inspected `reports/deduplication_audit.csv`, confirmed that the 91 removed observations were older filing contexts for the same metric/year/unit, and checked `groups_with_conflicting_values_across_filings` in the audit report. It was 0. I also ran the uniqueness/count test, which confirmed 152 source observations minus 91 older contexts equals 61 unique output rows. **Before submitting, the student should personally repeat this check and rewrite this paragraph in their own words.**

Sample selection

`data/sample/spotify_clean_sample_50.csv` contains the most recent 50 cleaned rows, sorted reproducibly by year and metric. This emphasizes decision-relevant recent performance while still including difficult cases: negative profit/loss, the 2022 low operating cash flow, instant cash records with blank start dates, and repeated-context selections visible through accession numbers. The full 61-row cleaned file and 152-row audit remain available for complete verification.